

Sheet 1 — Complex numbers

1. Compute

- a) $(2 + 3i) + (-1 - 4i)$
- b) $\left(-\frac{1}{2} + i\right) + \left(\frac{1}{2} - i\right)$
- c) $(2 - i) \cdot (1 + 2i)$
- d) $(-1 + i) \cdot (-1 - i)$

2. Find the complex number z that satisfies the condition $\begin{cases} \operatorname{Re}(z) + \operatorname{Im}(z) = 12 \\ \operatorname{Im}(z) = 3 \operatorname{Re}(z) \end{cases}$

3. Given $z_1 = 3 - 2i$ and $z_2 = (\alpha - 2) + (\beta + 3)i$, with $\alpha, \beta \in \mathbb{R}$, find α and β such that $z_1 + z_2$ is real and $z_1 \cdot z_2$ is purely imaginary.

4. Compute the modulus of each number: a) $3 - 4i$ b) $1 + i$ c) $3i$ d) 4

5. Given the complex numbers $z_1 = 3 - 2i$ and $z_2 = -4 + i$, check that:

- a) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
- b) $z_1 + \overline{z_1} = 2 \operatorname{Re}(z_1)$
- c) $z_2 - \overline{z_2} = i \cdot 2 \operatorname{Im}(z_2)$

6. Given that $z \cdot \bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2 = |\bar{z}|^2$, write as a product of complex numbers:

- a) $a^2 + b^2$ b) $x^2 + 4y^2$

7. Compute a) i^{37} b) i^{1990} c) i^{40} d) $(3i^{27} + i^{40})(2i^{61} - i^{18})$

8. Write in the form $a + bi$, with $a, b \in \mathbb{R}$: a) $\frac{3 - 2i}{1 + i}$ b) $\frac{5 - 2i}{4 + i}$

9. Solve the following equations in $\mathbb{C}$: a) $4x^2 - 4x + 5 = 0$ b) $x^3 + x^2 + 4x + 4 = 0$

10. Write in trigonometric form:

- a) $-3i$ b) 2 c) -1 d) $4i$ e) $1 + i$ f) $-\sqrt{3} + 3i$ g) $-2 - 2i$ h) $1 - i\sqrt{3}$

11. Write in algebraic form: a) $2 \operatorname{cis} \frac{2\pi}{3}$ b) $3 \operatorname{cis} 180^\circ$ c) $\operatorname{cis} \left(-\frac{\pi}{4}\right)$

12. With $z = x + yi$, identify in the Argand diagram the sets of points defined by the conditions:

- a) $|z| = 3$ b) $|z + 2| = 1$ c) $|z + i| = 2$ d) $|z - 1 + 2i| = 1$
- e) $|z - 3i| \leq 1$ f) $|z - 2 - i| > 9$ g) $1 \leq |z - 1 + 2i| \leq 4$

13. With $\bar{z}$ the conjugate of $z = x + yi$, identify in the Argand diagram the points that satisfy the condition $z \cdot \bar{z} = \bar{z} + z$.

14. Use De Moivre's formulae to solve the equation $z^4 - 8z = 0$.